

# Quiz 7: Hair & DNA Evidence

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*Review of Session 7. ~15 minutes.*

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## Section A: Hair structure & species

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1. Name the three layers of a hair, innermost to outermost.
2. Which cuticle scale pattern belongs to a cat? To a human?
3. Is a human medulla usually continuous or fragmented/absent? An animal medulla?
4. What is hair made of? What pigment gives it color?

## Section B: The growth cycle

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1. List the four growth phases in order.
2. Which phase are ~85–90% of scalp hairs in, and how long does it last?
3. A hair with a hard club-shaped root and no tissue — was it pulled or shed? What does that mean for DNA?
4. About how fast does human scalp hair grow (per month or per year)?

## Section C: Medullary index & the classic questions

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1. Write the medullary index formula.
2. Human MI is less than \_\_\_\_\_. Animal MI is more than \_\_\_\_\_.
3. Medulla 14  $\mu\text{m}$ , shaft 56  $\mu\text{m}$ . MI? Human or animal?
4. Does hair keep growing after death? Explain.
5. Hair's absorbent property makes it important for detecting what kind of poisoning, and why?

## Section D: DNA gel

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1. DNA has what charge, so it moves toward which electrode?
  2. Do smaller or larger fragments travel farther?
  3. How do you find which suspect matches the crime-scene DNA on a gel?
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## Answer Key

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### Section A

1. Medulla, cortex, cuticle.
2. Cat = spinous (petal-like); human = imbricate (flat, overlapping).
3. Human = fragmented or absent; animal = continuous (and wide).
4. Keratin; melanin.

### Section B

1. Anagen, catagen, telogen, exogen.
2. Anagen; 2–7 years.
3. Shed (telogen); poor for nuclear DNA — only mitochondrial DNA.
4. About 1 cm per month ( $\sim\frac{1}{2}$  inch/month,  $\sim 6$  inches/year).

### Section C

1.  $MI = \text{medulla diameter} \div \text{hair-shaft diameter}$ .
2. Less than  $\frac{1}{3}$  ( $\approx 0.33$ ); more than  $\frac{1}{2}$  ( $\approx 0.5$ ).
3.  $14/56 = 0.25 \rightarrow \text{human}$ .
4. No — the skin dehydrates and retracts, pulling back from the hair bases so they *appear* longer. No new growth.
5. Arsenic (heavy-metal) poisoning — the metal binds to keratin and stays fixed in the hair, preserving a datable timeline of exposure.

### Section D

1. Negative charge  $\rightarrow$  moves toward the positive electrode.
2. Smaller fragments travel farther.
3. Line up the crime-scene lane against each suspect lane; the suspect whose bands sit at the same heights is the match.